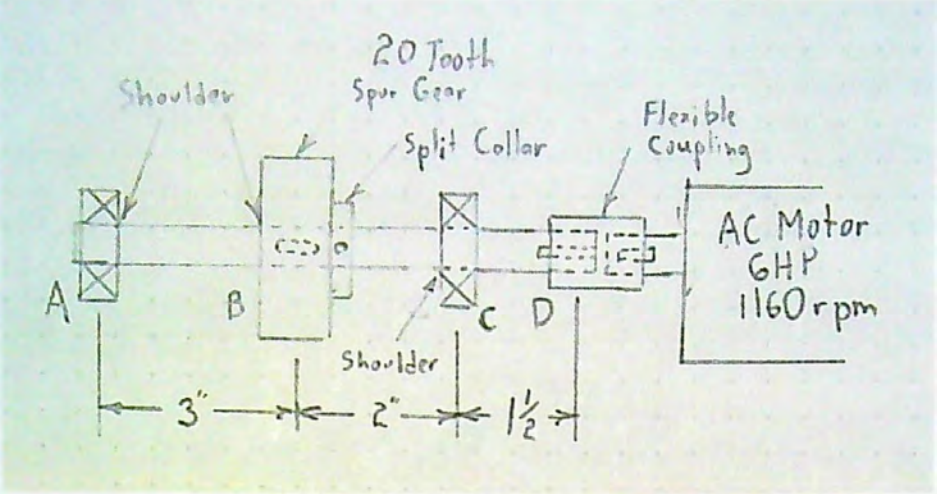
Scenario 1 – Shafts and Bearings (25 marks)

We are now expected to design the shaft to support the spur gear that we analyzed in the midterm. The shaft is supported by two **deep groove** ball bearings at A and C. The 20 tooth spur gear (pitch = 12 teeth/inch, $\phi = 20°$) is located at position B.



Proposed Shaft Layout (Note: Shaft Diameters to be Determine)

The spur gear is supported by a **large radius shoulder** on the left and a split collar (stress concentration factor, $K_t = 1.0$) on the right. The two bearings are press fit up against **sharp fillet shoulders** at A and C. Recall from the midterm the following forces were calculated for the 20 tooth spur gear (20° Pressure Angle).

Face Width	1.0 inch
Transverse Force W_t	391 lbf
Radial Force W_r	142 lbf
Rotational Speed	1160 rpm
Power	6 HP

A **profile keyseat**, located at B, and a **sled runner keyseat**, located at D, are used to transfer the torque.

a) Determine the reactions at A and C and draw the shear, bending moment, and torque diagrams for the shaft. Be sure to label the peak values. (5 marks)

b) The shaft is to be **machined** using SAE 1050 Cold Drawn Steel ($S_u = 100$ ksi, $S_y = 84$ ksi). Using a desired reliability of 99% and an initial estimate for the shaft diameter as 1.5 inch, estimate the endurance strength for the shaft (Note: treat all other Marin factors as 1.0). (3 marks)

c) Determine the required diameter of the shaft at the location of the spur gear (Point B). Use the standard design factor, N= 2.5. Determine the minimum required key length for the spur gear if the key is to be made from SAE 1018 Steel. (7 marks)

d) Size the required deep groove ball bearings for locations A and C. Use the same bore for the bearings at A and C in order to simplify the shaft design. Recall from the midterm that the machine was to operate for 6 hours/day, 20 days/month for 5 years. Use a reliability factor of 90%. Specify the required diameter of the shaft at A and C for the bearing along with the minimum shoulder diameter. (7 marks)

e) Propose a layout for the shaft including the shaft diameters. (3 marks)

Question 1

a) Reactions at A & C

Total force at B $= W_T = [W_T^2 + W_R^2]^{1/2} = (391^2 + 142^2)^{1/2}$

$W_T = 416$ lbf

$\Sigma M_A = 0 \quad (416)(3'') + 5''(R_C) = 0 \Rightarrow R_C = 250$ lbs (2 mk)

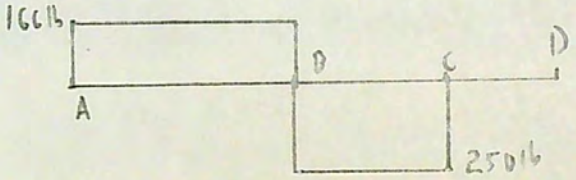
$\Sigma F = 0 \quad 250 - 416 + R_A = 0 \Rightarrow R_A = 166$ lbs

Shaft torque $T_{in} = 63000$ Power(HP)/n(rpm) Eg'n 12-1

$T = 63000\,(6\text{HP})/1160\text{rpm}$ (1 mk)

$T = 326$ in·lb

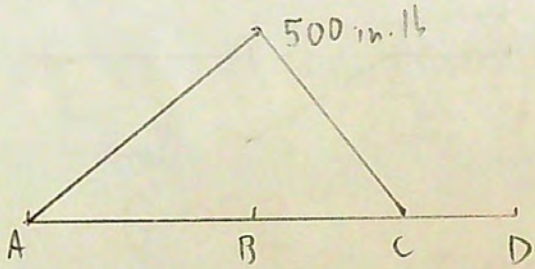
Shear (5mk)



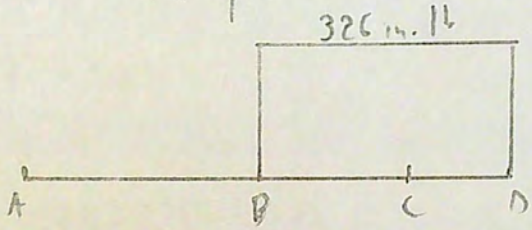
Moment

$BM_{max} = (166^{\#})(3'')$

$= 500$ in-lb



Torque



Torque is delivered @ D and exits @ B (3mk)

b) Se for shaft SAE 1050 CW [$S_u = 100$ ksi $S_y = 84$ ksi]

modified strength $= S'_e = S_n(C_m)(C_{st})(C_R)(C_s)$ Eg'n 5-21 (3mk)

$C_m = C_{st} = 1$

$S_n = \sim 38000$ psi Figure 5-11 machined steel @ $S_u = 100$ ksi

Reliability $C_R = 0.81$ Table 5-3 for $R = 99\%$

Size Factor $C_s = 0.85$ Figure 5-12 for $D \simeq 1.5$ inches [initial guess]

$\therefore S'_e = (38000)(0.81)(0.85)$

$\boxed{S'_e = 26163 \text{ psi}}$

Question 1 cont.

c) Required Dia. at Spur Gear B

i) We have steady torque and alternating bending moment

Design Equation for Shaft

1 mk $$D = \left[\frac{32N}{\pi} \sqrt{\left(\frac{K_f M}{S'_n}\right)^2 + \frac{3}{4}\left(\frac{T}{S_y}\right)^2} \right]^{1/3}$$ Eq'n 12-24

For spur gear we have large shoulder on left side $K_T = 1.5$

we have profile keyseat for torque delivery $K_T = 2.0$

1 mk we have split collar on right side $K_T = 1.0$

∴ keyseat dominates $K_T = 2.0$

@ B Bending Moment $M = 500$ in-lb & $T = 326$ in-lb

7 mk $S'_e = 26163$ psi & $S_y = 84000$ psi

Design factor $N = 2.5$ (given)

2 mk ∴ $$D = \left[\frac{32(2.5)}{\pi} \sqrt{\left(\frac{(2.0)(500)}{26163}\right)^2 + \frac{3}{4}\left(\frac{326}{84000}\right)^2} \right]^{1/3}$$

$D = 0.992''$

1 mk Select Preferred Size [Table A2-1]

∴ $\boxed{D = 1.000 \text{ inches}}$

ii) Minimum required key size

1 mk $L = \frac{4TN}{DWS_y}$ Eq'n 11-5 [key material is weakest]

For 1" Dia. shaft $W = H = 0.2500$ [Table 11-1]

For 1018 Carbon steel key $S_y = 54000$ psi [Table 11-4]

1 mk ∴ $L = \frac{4(326)2.5}{(1.00)(0.25)(54000)} = \boxed{0.241''}$

For 1" spur gear we can select 1/2, 5/8 or 3/4" key

Note: eq'n uses Tresca Criteria

$S_{sy} = 0.5 S_y$

$N = 2.5$ (given)

Question 1 (cont.)

d) Size the bearings for A & C

$$\text{lifetime} = (1160\,\text{rpm})\left(\frac{60\,\text{min}}{1\,\text{hr}}\right)\left(\frac{6\,\text{hr}}{\text{day}}\right)\left(\frac{20\,\text{day}}{\text{month}}\right)\left(\frac{12\,\text{mo}}{1\,\text{yr}}\right)(5\,\text{year})$$

(1 mk) $L_2 = 501 \times 10^6$ cycles

Bearing Load & Life relationship

$$L_2/L_1 = \left(P_1/P_2\right)^k \quad \text{[Eqn 14-1]}$$

For ball bearings $k = 3.00$

Load at C = 250 lbs = reaction @ C = P_2

$P_1 = C_{10}$ rating @ 1×10^6 load cycles = L_1

(1 mk) $\therefore \quad 501\times10^6/1\times10^6 = \left(C_{10}/250\,\text{lbs}\right)^3$

(1 mk) $C_{10} = 250(501)^{1/3} = 1985$ lbs

Min. shaft dia @ C

$$D = \left[\frac{32N}{\pi}\sqrt{\left(\frac{KM}{S'_n}\right)^2 + \frac{3}{4}\left(\frac{T}{S_y}\right)^2}\right]^{1/3} \quad \text{eq'n 12-24}$$

(1 mk) Here M = D and T = 326 lb·in [part a]

$$\therefore D_{min} = \left[\frac{32(2.5)}{\pi}\sqrt{\frac{3}{4}\left(\frac{326}{84000}\right)^2}\right]^{1/3} = 0.441 \text{ inches}$$

We need to also check for alternating shear

[eq'n 12-16] $$D_{min} = \sqrt{\frac{2.94\,K_T(V)N}{S'_n}}$$

$K_T = 2.5$ sharp shoulder

$V = 250$ lbs

$N = 2.5$ (given)

$S'_n = 26163$ psi (part b)

(1 mk)
$$D_{min} = \sqrt{\frac{2.94(2.5)(250)(2.5)}{26163}}$$

$D_{min} = 0.419$ inches

so $D_{min} = 0.441$ inches $= (0.441)(25.4\,\text{mm/in.}) = 11.20$ mm. (min)

$\therefore$ bearing bore exceed 11.2 mm.

Problem 1 d) cont.

(1 pt) Using Table 14-3 $C_{10} > 1985$ lbs, and $D > 11.2$ mm

Bearing C **Select 6301 Deep Groove Ball Bearing, Bore = 12mm**

$C_{10} = 2192$ lbs, min fillet radius = 1.0 mm & min shoulder dia $D_{S_{min}} = 17$ mm.

Bearing A $L_2/L_1 = (P_1/P_2)^k$ Load @ A = 166 lbs = P_2

(1 pt) $\therefore C_{10}$ @ A = $166(501)^{1/3} = 1318$ lbs

We want bore @ A = bore @ C = 12mm

We can select a **6201 Deep Groove Ball Bearing Bore = 12mm**

D_S @ A = 16mm

(1 pt) **Final Dia @ A & C = $0.4723^{+0.0003}_{-0.0000}$** Table 15-5 for j5 ISO Tolerance Fit

e) Dia. @ D [Flexible Coupling]

(1 pt) At D we only have steady torque T = 326 lb·in

From part c) $D_{min} = 0.441$ inches [eq'n 12-24 No BM]

If we want a recommended Basic Size we must select $\frac{1}{2}'' = D_D$

(2 pts) Then Bearing C must be 15 mm 6302 $C_{10} = 2563^{\#}$ $D_{S_{min}} = 20$mm

Bearing A must be 15 mm 6202 $C_{10} = 1754^{\#}$ $D_{S_{min}} = 19$mm.

Min. keyway length at D $L_{min} = \frac{4(326)(2.5)}{(0.5)(0.125)(54000)} = 0.966 = 1''$

(3 pts) Final Dia @ A & C = $0.5905^{+0.0003}_{-0.0000}$

Scenario 2 (15 marks)

Fluid Systems

A hydraulic press is required to bend steel plates during the construction of a ship. The press must provide **a maximum of 4000 lbs of load** on the plate. The working pressure of the system is **1000 psi** and the piston rod must extend a **maximum of 8 inches**. The hydraulic cylinder is rigidly mounted vertically with the rod extending in the downward direction. The end of the rod is not guided at all.

a) Size a cylinder and rod for this operation. (5 marks)
b) If we wish to bend the plate at a rate of 3 inches per second, determine the flow rate required by the pump. What diameter of piping is required and what size of motor is required? Do you expect acceleration will change your results? Explain. (5 marks)
c) Sketch a hydraulic circuit for the <u>manually</u> operated press. Is a regenerative circuit suitable for this application?

a) Table b-1 $D = 2\frac{1}{2}"$ $F_{max} = 4910$ lb @ 1000 psi

(2 mk) stroke = 8 inches

(5) cylinder rigidly supported but end is free ∴ Stroke Factor = 2.0 [Table b-4]

(2 mk) ∴ basic length = 2 × 8 = 16 inches

For thrust = 4000 lb @ 16" basic length $d_{rod} = 1"$

(1 mk) round up in graph b-1

∴ **$2\frac{1}{2}$ bore & 1" × 8" stroke rod**

b) extension speed = 3"/sec = 15 ft/min

(1 mk) Using table b-5 $2\frac{1}{2}$" bore & 10" piston rod 2.55 gpm for 10 ft/min [Table b-5]

(1 mk) ∴ for 15 ft/min we need 1.5 × 2.55 = **3.83 gpm = Q**

(5) keep V < 15 ft/s in piping select 3/8" dia = 5.80 ft/sec @ 10 ft/min

(1 mk) $V_{pipe} = 1.5(5.80) = 8.7$ ft/sec < 10 < 15 **3/8 pipe**

(1 mk) HP = GPM × PSI / 1714 (0.85) = (3.83)(1000)/1714(0.85)

$HP_{min} = 2.63$ HP

(1 mk) Nearest Motor = 3HP either single phase or 3 phase

(over)

b) cont.

Using graph b-2 we see that for a $V_{max} = 15$ ft/min the maximum acc. force factor = ~ 0.05g [S = ¼"]

(1mk)

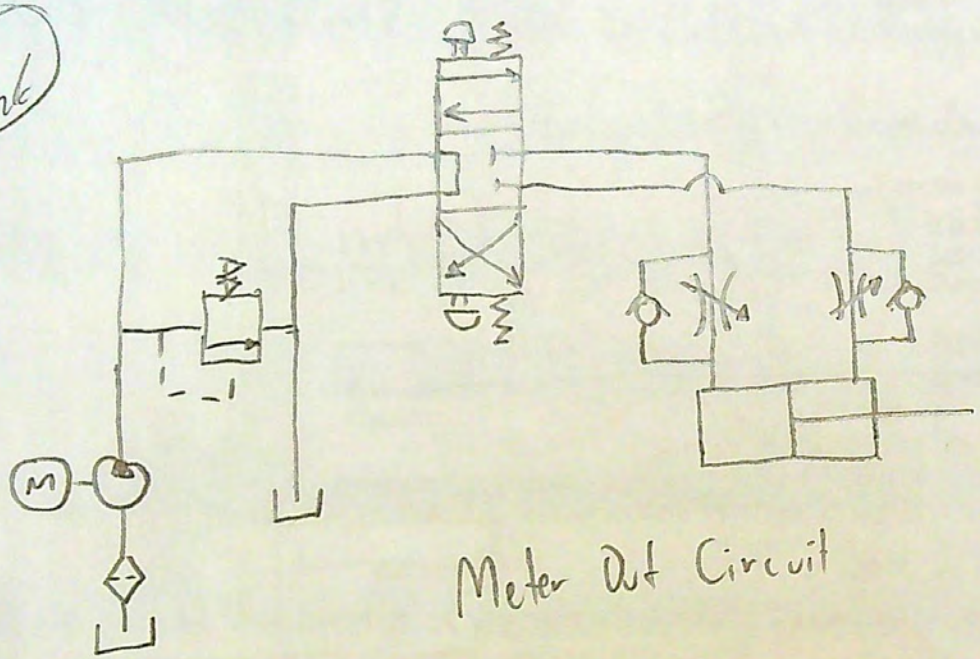
This is only a 5% increase in thrust

Thrust max = 4000 + 5% = 4200 lbs.

This is still below 4910 lb rating so no issue with acceleration

c) Numerous choice

(3mk)



Meter Out Circuit

(2mk)

Regenerative circuit is not ideal in this case as rod dia. must be 2½". Min cylinder size is a 4" bore

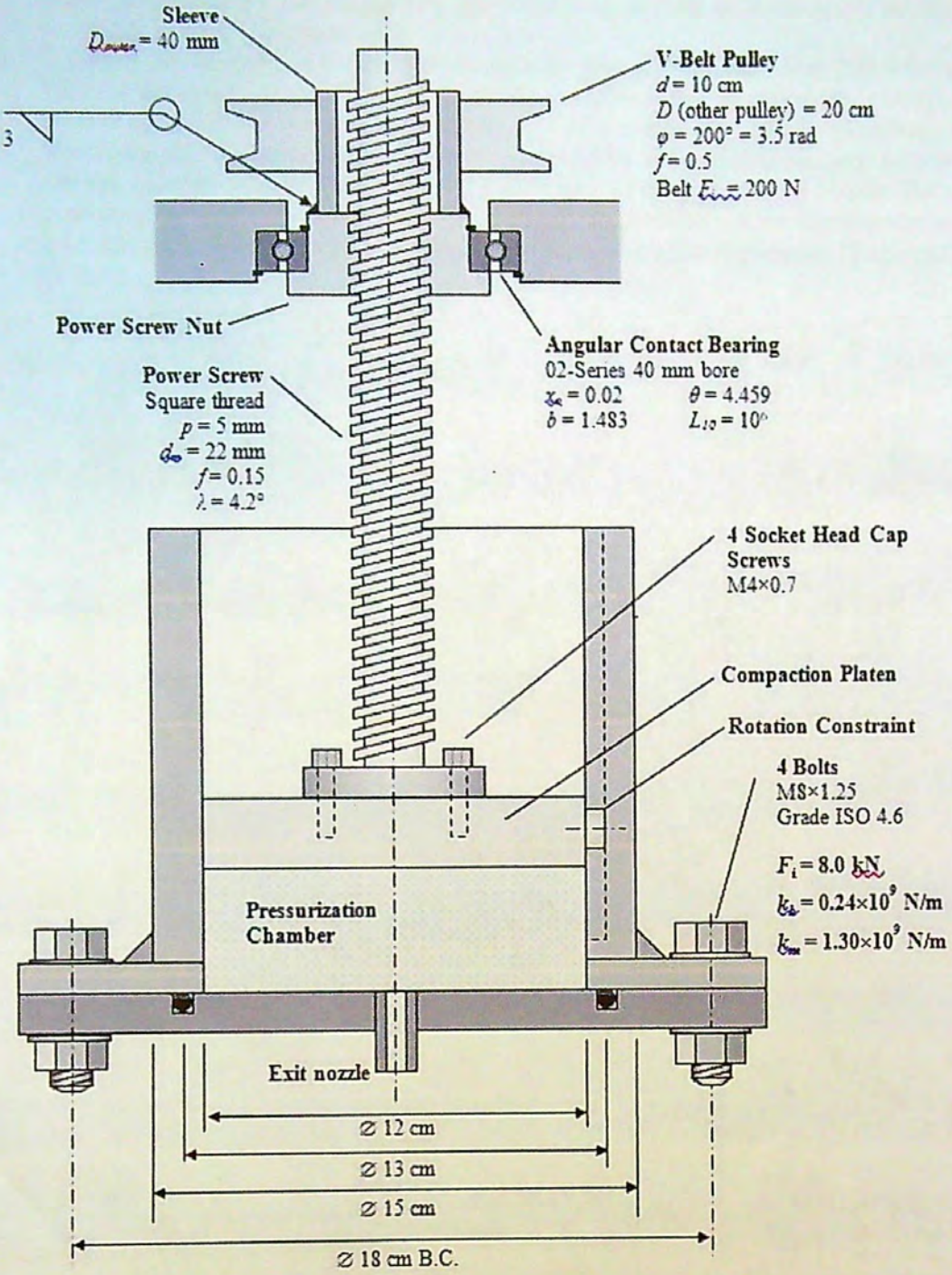
here $q_{piston} = 6.53$ gpm and $q_{rod} = 3.98$ gpm for 10 ft/min

$\therefore q_{pump} = 6.53 - 3.98 = 2.55$ gpm

The same or 2½ bore cyl. so no savings in pump size

Scenario 3 (20 marks)

Fasteners and Power Screws



An assembly-line device to repeatedly provide a pressurized supply of a thick liquid-gel is shown above. For this design, a V-Belt connects the motor to the power screw. The pressurization chamber has a circular cross-section with **12 cm inside diameter** and a small exit nozzle on the bottom. During use, pressure in chamber, plus wall friction, can be assumed to create a **maximum axial Force of 2400 Newtons**. The pressure force alternates between 2400 Newtons and 0 Newtons ten times a minute. The raising force during the return stroke is negligible.

The bottom plate is held by **four** M8×1.25 bolts (Grade ISO 4.6) evenly spaced on an 18 cm base circle. Standard nuts and flat washers are used. The O-ring can be ignored.

NOTE: Bolt preload (F_i = 8.0 kN) and the calculated spring rates (k_b = 0.24×10^9 N/m, k_m = 1.30×10^9 N/m) for the bolts and the members are shown on the figure (bottom right).

1. What fraction of the load is carried by the bolts? Find the yield factor of safety for the bolts used to secure the bottom plate. (8 points)
2. Knowing that the modified endurance strength for the rolled threads of the ISO 4.6 bolts is S_e = 48 MPa., sketch the fatigue failure diagram for this condition. Use the Goodman Criteria for Failure. What is the safety factor for fatigue? Comment on the results of 1) and 2). (6 points)
3. Determine the torque required to drive the power screw at the maximum load. Assume the friction in the ball bearings is negligible. What is the efficiency of the power screw? **Note**: The information regarding the square thread power screw is shown in the middle of the figure (mean screw diameter d_m = 22 mm.). Would you suggest a ball screw be used for this application? Explain. (6 points)

a) (2 mk) $C = \frac{k_b}{k_b + k_m} = \frac{0.24\times10^9}{(0.24+1.3)\times10^9} = 0.156 = 15.6\%$ carried by bolt

(1 mk) Load/bolt = 2400/4 = 600 N F_i = 8000 N

From Table 19-5 or 8-1 $A_t = 36.6\times10^{-6}\ m^2$

(1 mk) Table 19-7 or 8-11 ISO 4.6 bolt S_p = 225 MPa Proof Strength
S_y = 240 MPa

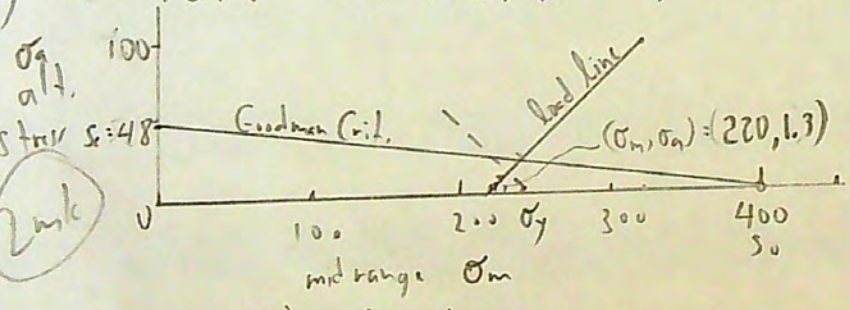
Stress on bolt at Full load (2 mk)

$$\sigma_b = \frac{C(P)+F_i}{A_t} = \frac{(0.156)(600)+8000}{36.6\times10^{-6}}$$

$$\sigma_b = 221.1\ MPa$$

(2 mk) ∴ safety factor yield [Proof Strength] $n_{s,F_y} = S_p/\sigma_b = \frac{225\ MPa}{221.1\ MPa} = \boxed{1.02 = n_y}$

b) S_e = 48 MPa S_u = 400 MPa S_y = 240 MPa



(2 mk)

$\sigma_a = \frac{1}{2}\left(\frac{CP}{A_t}\right) = \frac{(0.156)600}{2(36.6\times10^{-6})} = 1.28\ MPa$

Very safe in fatigue due to high preload. Safe design. (1 mk)

(1 mk) $\sigma_i = F_i/A_t = 8000/36.6\times10^{-6} = 219\ MPa$

(2 mk) $n_f = \frac{S_e(S_u-\sigma_i)}{\sigma_a(S_u+S_e)} = \frac{48(400-219)}{(1.28)(400+48)}$ Mod. Goodman Failure Criteria [Eq. 8-45]

$n_f = 15.2$ very safe!

$\boxed{n_f = 15}$

(over)

c) Torque for Power screw

To push screw down is the same as lifting the collar

[Eq'n 17-2] Lifting force for square thread

(1mk) $$T = \frac{F D_p}{2}\left[\frac{L + \pi f D_p}{\pi D_p - fL}\right] + \text{Collar Friction}$$

With roller bearings assume collar friction = zero

We have single thread so $p = L = 5\text{ mm}$. $D_p = 22\text{ mm}$.

F = total load = 2400 N f = thread friction = 0.15

$$\therefore T = \frac{(2400\text{N})(22\times10^{-3}\text{m})}{2}\left[\frac{(5\times10^{-3}\text{m}) + \pi\, 0.15\,(22\times10^{-3}\text{m})}{\pi(22\times10^{-3}\text{m}) - 0.15(5\times10^{-3}\text{m})}\right]$$

(2mk) $$= (26.4)\,\frac{0.01537}{0.0684}$$

$$T = 5.93\text{ N.m} = 4.36\text{ ft.lb}$$

$$\text{Eff} = \eta = \frac{T'}{T_u} = \frac{FL}{2\pi T_u} = \frac{(2400)(5\times10^{-3})}{2\pi(5.93\text{ N.m})}$$

(2mk) $$\boxed{\eta = 32.2\%}$$

(1mk) A ball screw would be more efficient & 2400N (539 lbs) is well within the range for ball screws